

Underwriting & Customer Analytics Dashboard

InsurTech & Digital Risk Solutions

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Date: 27/08/2026

1. Project Overview

This project uses the **Underwriting & Customer Analytics Dashboard** — built on the course-provided synthetic datasets (customers.csv, policies.csv, and claims.csv, seed 42). The dashboard turns the insurer's policy book into a single view of who the book is insuring, how it is being underwritten, and where early risk signals are emerging.

Business problem: underwriting and portfolio-risk teams currently see premium, credit quality, and KYC compliance as separate reports. This dashboard consolidates them so a single view can answer what the risk profile of the active book looks like today, which segments are driving claims incidence, and where compliance exposure is concentrated.

Intended audience: the Head of Underwriting or Chief Risk Officer at a mid-sized Indian general insurer, who needs a fast, evidence-based read on portfolio risk to guide underwriting policy, channel strategy, and KYC remediation priorities — without waiting on a manual cross-team report.

2. Data Preparation

Sources

customers.csv (5,000 rows), policies.csv (10,000 rows), and claims.csv (20,000 rows) — synthetic, course-provided, fixed seed 42. No real/external data has been added in this cut of the dashboard.

Joins

policies is left-joined to customers on customer_id to build the base analysis table (df). claims is aggregated to policy level first — claims_count (count of claim_id) and total_claim amount (sum of claim_amount) per policy_id — then left-joined onto df, since premium is a policy-level field and a single policy can carry several claims. Policies with no matching claims are filled to 0 rather than left null, since “zero claims” is a valid, meaningful state for this book.

Derived columns

start_date and claim_date are parsed to datetime. start_month buckets start_date to a year-month period for time-series grouping. has_claim is a boolean flag (claims_count > 0). credit_band splits credit_score into four equal-sized quartiles (Q1-Low → Q4-High) using pd.qcut, so each band represents the same share of the book rather than a fixed score range.

Data-quality notes and assumptions

Approximately 5.0% of customer income values are missing. Rather than imputing a value, income-based views simply exclude these records — imputing here would manufacture precision the data doesn't support.

Loss ratio is deliberately not used as a headline KPI. The synthetic premium field is not calibrated to claims cost, which produces loss-ratio values above 2,000% — not interpretable as a real underwriting signal. This dashboard's profitability view is built on claims-side metrics instead (frequency, incidence by segment),

consistent with the limitation the assignment brief itself flags for this dataset.

3. KPI Definitions

Five KPIs are used, computed on the joined analysis table (df); two are calculated/derived metrics rather than raw counts.

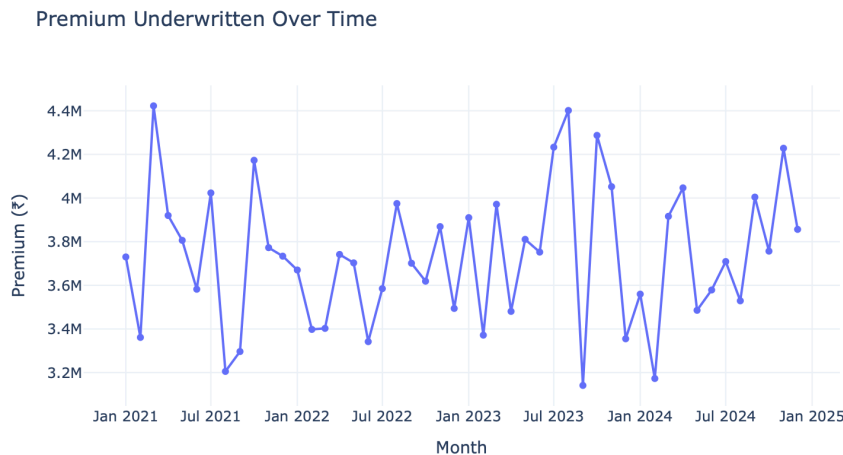
KPI	Formula	Why it matters
Total Premium Underwritten	$\Sigma \text{ premium across policies in view}$	Headline measure of book size — the base every other underwriting metric is read against.
Avg Premium / Policy	$\Sigma \text{ premium} \div \text{count}(\text{policies})$	Shows whether the book is growing through more policies or richer ones; a downward drift against sum assured flags underpricing.
Avg Sum Assured	$\Sigma \text{ sum_assured} \div \text{count}(\text{policies})$	Proxy for capital at risk per contract, independent of premium collected.
Claims Frequency Rate (derived)	$(\Sigma \text{ claims_count} \div \text{count}(\text{policies})) \times 100$	Derived metric: claims volume relative to book size. Values above 100% mean more than one claim per policy on average and should trigger review of claims-prone segments.
KYC Verified Rate (derived)	$\text{count}(\text{kyc_status} = \text{'Verified'}) \div \text{count}(\text{policies}) \times 100$	Derived compliance metric — a low rate signals AML/regulatory exposure and renewal-time friction.

4. Dashboard Tour

Insert a screenshot of each visual here once the final interactive dashboard is exported.

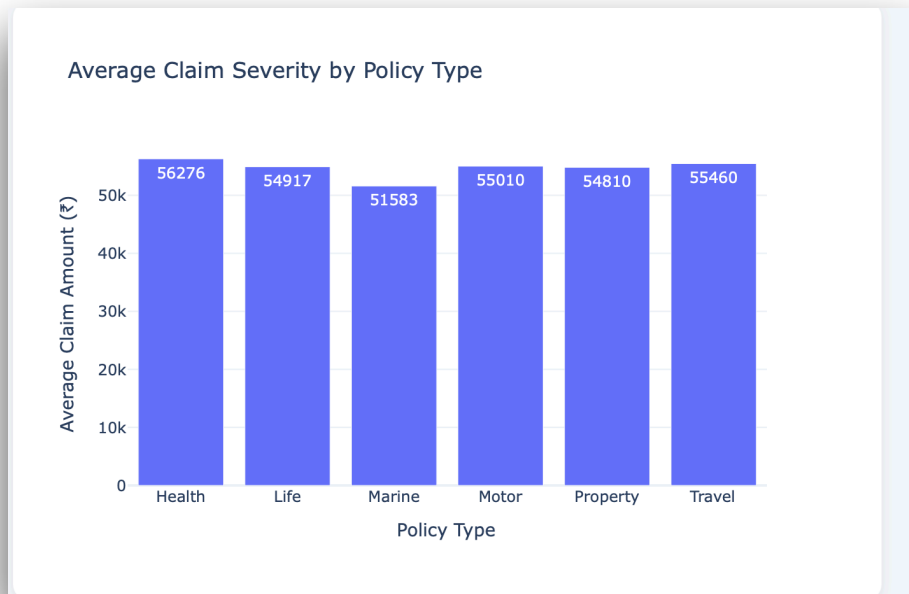
1. Premium Underwritten Over Time (line, Jan 2021 – Dec 2024)

Monthly premium underwritten moves in a fairly tight band — from ₹31.4 lakh (Sep 2023) to ₹44.2 lakh (Mar 2021) — with no sustained upward or downward trend across four years. Takeaway: book growth has plateaued rather than compounded, which is worth flagging if leadership's plan assumed expansion.



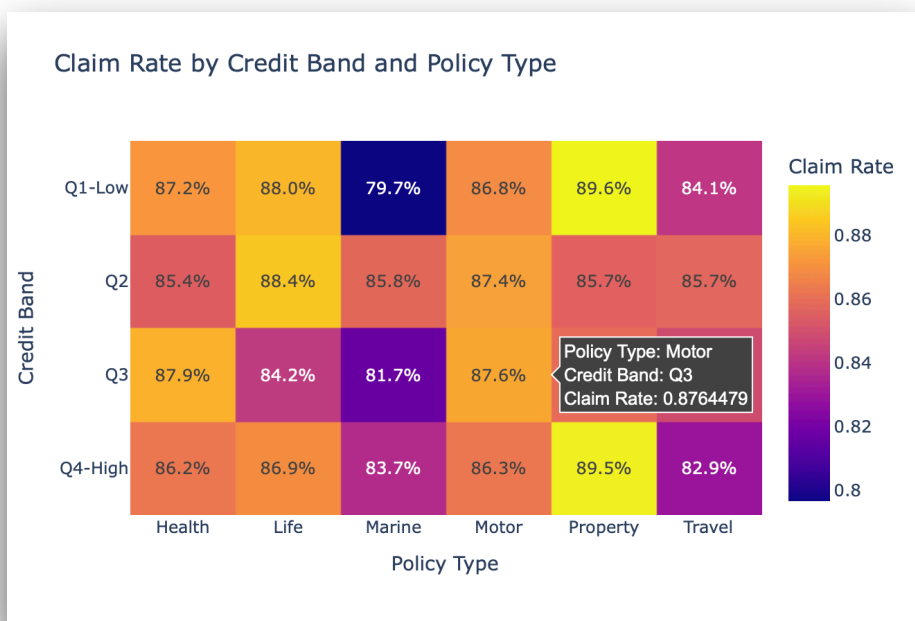
2. Average Premium by Channel (bar)

Broker writes the highest average premium (₹18,169), narrowly ahead of Online (₹18,064) and Bancassurance (₹17,968); Agent is lowest at ₹17,672. The spread across all five channels is under 3%, meaning channel choice alone barely moves average ticket size.



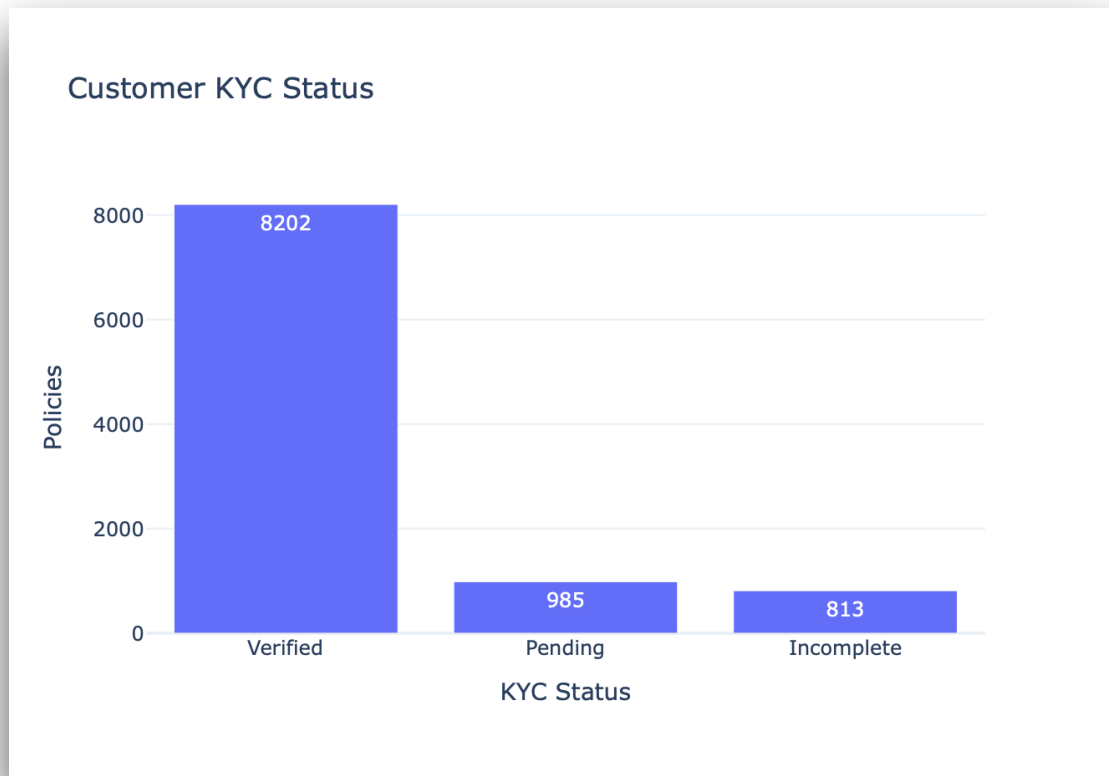
3. Claims Rate Heatmap — Credit Band × Policy Type

The standout finding on this dashboard: claim incidence is nearly flat across credit-score quartiles (85.4%–86.4%, a sub-1-point spread), but varies meaningfully by product line — from 82.7% on Marine to 87.7% on Property, a 5-point spread. In this book, credit score is not currently separating good risk from bad; product line is.



4. KYC Status Funnel

8,202 policies (82.0%) sit on Verified KYC; 985 (9.9%) are Pending; 813 (8.1%) are Incomplete. Nearly one in five active policies — 18.0% — carries unresolved KYC status, a compliance exposure sitting inside the live book, not just at onboarding.



5. Insights & Recommendations

1. Re-weight underwriting risk tiering away from credit score, toward product line.

Evidence: Claims-rate heatmap: credit-band spread is under 1 point (85.4%–86.4%) vs a 5-point spread by product line (Marine 82.7% low, Property 87.7% high).

Expected impact: Sharper risk segmentation with no new data required; scrutiny can move to Marine and Travel, where incidence diverges most, instead of a credit-based tiering that isn't currently discriminating risk in this book.

2. Run a targeted KYC remediation drive for the 1,798 Pending/Incomplete policies (18.0% of the book), prioritized by upcoming renewal date.

Evidence: KYC status funnel.

Expected impact: Directly reduces AML/compliance exposure ahead of the next audit cycle and cuts renewal friction for affected policyholders.

3. Investigate the four-year plateau in premium underwritten (₹31.4L–₹44.2L monthly, no clear trend) against the insurer's stated growth targets.

Evidence: Premium-trend line.

Expected impact: Establishes whether flat volumes reflect a deliberate strategy or lost distribution momentum — a prerequisite before setting next year's underwriting targets.

4. Improve income capture at point-of-sale (currently ~5% missing) before using income as an affordability or pricing input.

Evidence: Data-quality check, Section 2.

Expected impact: A complete income field would let future underwriting rules use affordability screening with confidence; today it can only be excluded from analysis.

6. AI Usage Log & Declaration

This project is AI Assessment Level 2. AI assistance was used for structuring this dashboard and walkthrough and for code/debugging help — the analysis, KPI values, chart data, and recommendations above are drawn directly from this project's own datasets and notebook outputs, not generated or invented by AI.

Per the assignment's own AI Usage Policy (Section 10), the log below and the declaration must be completed and signed personally — an AI tool must not write its own usage log. Replace each bracketed row with your actual prompts, tools, and verification steps before submission.

Section	What I asked (prompt)	Tool used	Where I used the output	How I verified it
1	Read my underwriting dashboard code and help me improve it to meet the assessment requirements and look like a professional Power BI-style dashboard.”	ChatGPT	Dashboard design and Python/Dash code	I checked the suggested dashboard structure against the assessment requirements for 5+ KPIs, 4+ visualizations, 2+ filters, and drill-down. I also checked that the proposed fields matched the columns in my notebook.
2	“My Jupyter Notebook gives an <code>__main__</code> has no attribute <code>__file__</code> error when running Dash/Plotly. Help me fix the error.	ChatGPT	Jupyter Notebook dashboard execution	I tested the suggested changes in Jupyter and identified that the error was related to running the Dash/Plotly application from a notebook environment rather than the underwriting calculations.

Declaration

For the final section of your AI log, use the declaration **required by the assessment brief**:

“I declare that all analysis, insights, and recommendations in this project are my own, produced by me from the course datasets. AI tools were used only as documented above, and I verified every AI-assisted output before use.”